

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print("Libraries Imported Successfully")
```

Libraries Imported Successfully

```
df = pd.read_csv('Superstore.csv')

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Segment	Country	City	State	Region	Product ID	Category	Category Name
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Consumer	United States	Henderson	Kentucky	South	FUR-BO-10001798	Furniture	Bookcase
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Consumer	United States	Henderson	Kentucky	South	FUR-CH-10000454	Furniture	Chair
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Corporate	United States	Los Angeles	California	West	OFF-LA-10000240	Office Supplies	Office Supplies
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Consumer	United States	Fort Lauderdale	Florida	South	FUR-TA-10000577	Furniture	Furniture
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Consumer	United States	Fort Lauderdale	Florida	South	OFF-ST-10000760	Office Supplies	Office Supplies

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 19 columns):
#   Column             Non-Null Count  Dtype
---  -
0   Row ID              9994 non-null   int64
1   Order ID            9994 non-null   object
2   Order Date          9994 non-null   object
3   Ship Date           9994 non-null   object
4   Ship Mode           9994 non-null   object
5   Customer ID         9994 non-null   object
6   Segment             9994 non-null   object
7   Country             9994 non-null   object
8   City                9994 non-null   object
9   State               9994 non-null   object
10  Region              9994 non-null   object
11  Product ID          9994 non-null   object
12  Category            9994 non-null   object
13  Sub-Category        9994 non-null   object
14  Product Name        9994 non-null   object
15  Sales               9994 non-null   float64
16  Quantity            9994 non-null   int64
17  Discount            9994 non-null   float64
18  Profit              9994 non-null   float64
dtypes: float64(3), int64(2), object(14)
memory usage: 1.4+ MB
```

```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Segment	0
Country	0
City	0
State	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

```
total_sales = df['Sales'].sum()

print("Total Sales =", round(total_sales,2))
```

Total Sales = 2297200.86

```
total_profit = df['Profit'].sum()

print("Total Profit =", round(total_profit,2))
```

Total Profit = 286397.02

```
category_sales = df.groupby('Category')['Sales'].sum()

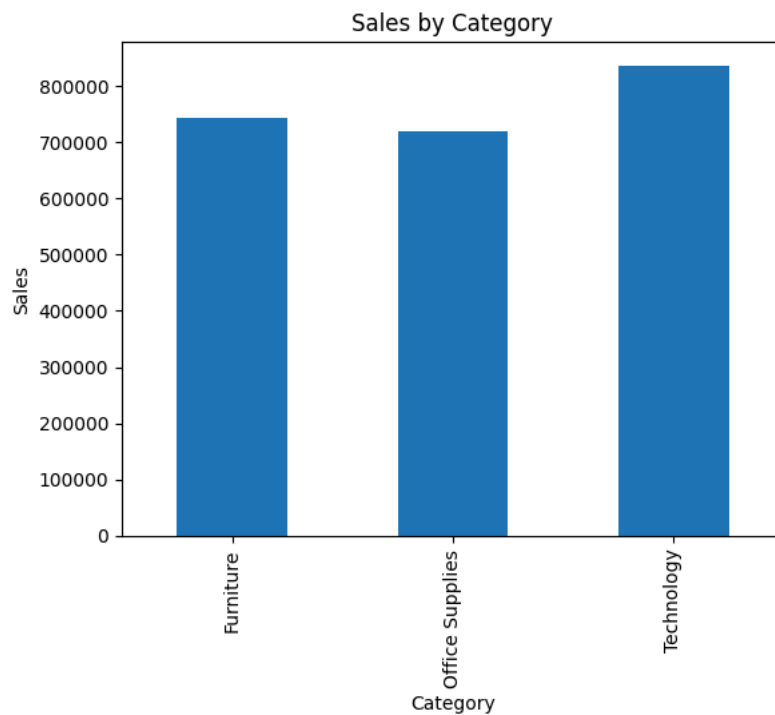
print(category_sales)
```

Category	
Furniture	741999.7953
Office Supplies	719047.0320
Technology	836154.0330

Name: Sales, dtype: float64

```
category_sales.plot(kind='bar')

plt.title("Sales by Category")
plt.ylabel("Sales")
plt.show()
```



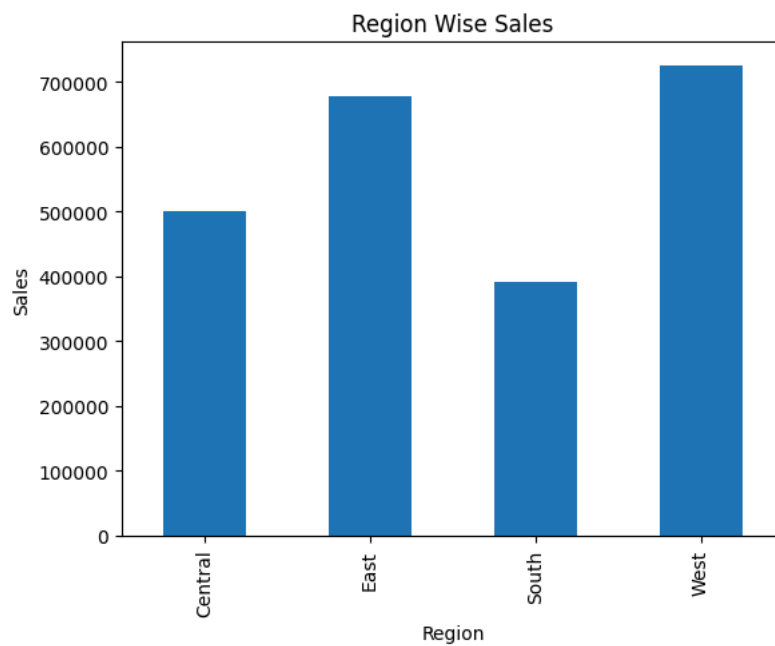
```
region_sales = df.groupby('Region')['Sales'].sum()

print(region_sales)
```

```
Region
Central    501239.8908
East       678781.2400
South      391721.9050
West       725457.8245
Name: Sales, dtype: float64
```

```
region_sales.plot(kind='bar')

plt.title("Region Wise Sales")
plt.ylabel("Sales")
plt.show()
```



```
top_products = df.groupby('Product Name')['Sales'].sum().sort_values(ascending=False).head(10)

print(top_products)
```

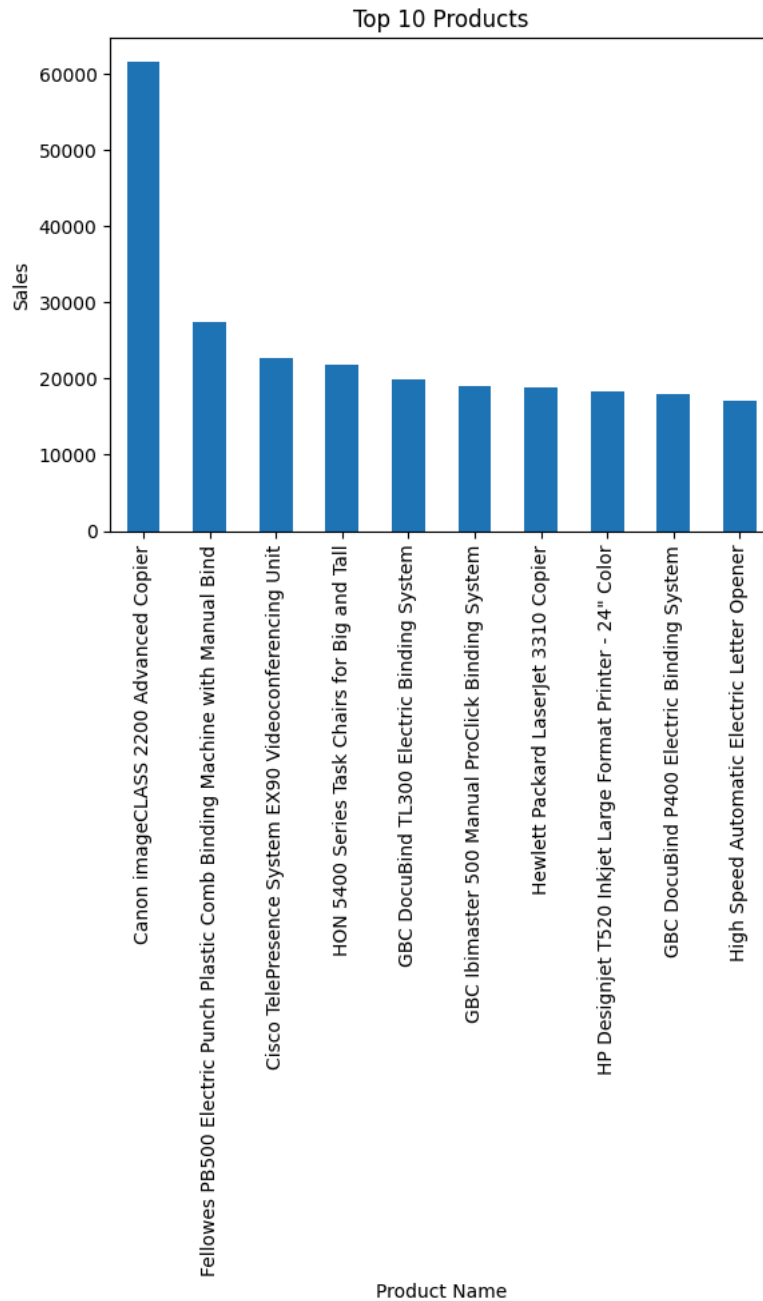
```
Product Name
Canon imageCLASS 2200 Advanced Copier          61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind  27453.384
```

Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

Name: Sales, dtype: float64

```
top_products.plot(kind='bar')
```

```
plt.title("Top 10 Products")
plt.ylabel("Sales")
plt.show()
```



```
segment_sales = df.groupby('Segment')['Sales'].sum()
```

```
print(segment_sales)
```

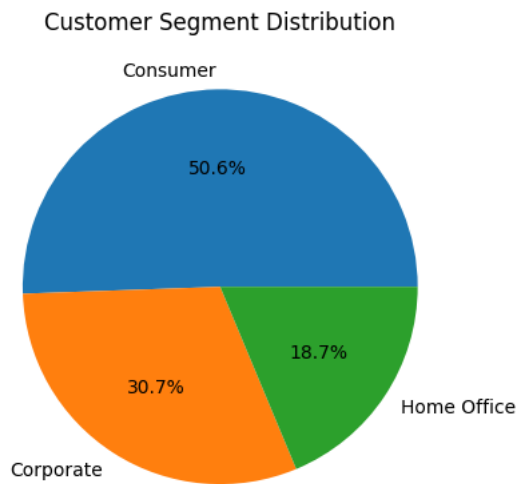
Segment	
Consumer	1.161401e+06
Corporate	7.061464e+05
Home Office	4.296531e+05

Name: Sales, dtype: float64

```
segment_sales.plot(kind='pie', autopct='%1.1f%%')
```

```
plt.title("Customer Segment Distribution")
```

```
plt.ylabel("")
plt.show()
```



```
profit_category = df.groupby('Category')['Profit'].sum()

print(profit_category)
```

```
Category
Furniture      18451.2728
Office Supplies 122490.8008
Technology     145454.9481
Name: Profit, dtype: float64
```

```
profit_category.plot(kind='bar')

plt.title("Profit by Category")
plt.ylabel("Profit")
plt.show()
```



```
correlation = df[['Sales', 'Profit', 'Quantity', 'Discount']].corr()

print(correlation)
```

```
Sales      Sales      Profit  Quantity  Discount
Sales      1.000000    0.479064  0.200795 -0.028190
...
```